

Age

	< 30	31-40	>40
1	N	Y	Y
2	N	Y	Y
8	N	Y	N
9	Y	Y	Y
11	Y		N

$$\text{Info}(D) = I(9,5) = \left(-\frac{9}{14} \log_2 \left(\frac{9}{14}\right) - \frac{5}{14} \log_2 \left(\frac{5}{14}\right)\right) = 0.940$$

$$\begin{aligned} \text{Info}_{\text{age}}(D) &= \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2) \\ &= \frac{5}{14} \left(-\frac{2}{5} \log_2 \left(\frac{2}{5}\right) - \frac{3}{5} \log_2 \left(\frac{3}{5}\right)\right) + \frac{4}{14} \left(-\frac{4}{4} \log_2 \left(\frac{4}{4}\right) - \frac{0}{4} \log_2 \left(\frac{0}{4}\right)\right) \\ &\quad + \frac{5}{14} \left(-\frac{3}{5} \log_2 \left(\frac{3}{5}\right) - \frac{2}{5} \log_2 \left(\frac{2}{5}\right)\right) \\ &= 0.347 + 0 + 0.347 \\ &= 0.694 \end{aligned}$$

$$\begin{aligned} \text{Gain}_{\text{age}} &= 0.940 - 0.694 \\ &= 0.246 \end{aligned}$$

Income

	low	medium	high
5	Y	Y	N
6	N	N	N
7	Y	Y	Y
9	Y	Y	Y
		12	Y
		14	N

$$\begin{aligned} \text{Info}_{\text{inc}}(D) &= \frac{4}{14} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2) \\ &= \frac{4}{14} \left(-\frac{3}{4} \log_2 \left(\frac{3}{4}\right) - \frac{1}{4} \log_2 \left(\frac{1}{4}\right)\right) \\ &\quad + \frac{6}{14} \left(-\frac{4}{6} \log_2 \left(\frac{4}{6}\right) - \frac{2}{6} \log_2 \left(\frac{2}{6}\right)\right) \\ &\quad + \frac{4}{14} \left(-\frac{2}{4} \log_2 \left(\frac{2}{4}\right) - \frac{2}{4} \log_2 \left(\frac{2}{4}\right)\right) \\ &= 0.231 + 0.394 + 0.286 \\ &= 0.911 \end{aligned}$$

$$\begin{aligned} \text{Gain}_{\text{income}} &= 0.940 - 0.911 \\ &= 0.029 \end{aligned}$$

	age	income	student	credit_rating	buys computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31...40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31...40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31...40	medium	no	excellent	yes
13	31...40	high	yes	fair	yes
14	>40	medium	no	excellent	no

student

	N	Y
1	N	Y
2	N	N
3	Y	Y
4	Y	Y
8	N	Y
12	Y	Y
14	N	Y

$$\begin{aligned} \text{info}_{\text{student}}(D) &= \frac{7}{14} I(3,4) + \frac{7}{14} I(6,1) \\ &= \frac{7}{14} \left(-\frac{3}{7} \log_2 \left(\frac{3}{7}\right) - \frac{4}{7} \log_2 \left(\frac{4}{7}\right)\right) + \frac{7}{14} \left(-\frac{6}{7} \log_2 \left(\frac{6}{7}\right) - \frac{1}{7} \log_2 \left(\frac{1}{7}\right)\right) \\ &= 0.493 + 0.296 \\ &= 0.789 \end{aligned}$$

Credit_rating

	fair	excellent
1	N	N
3	Y	N
4	Y	Y
5	Y	Y
8	N	Y
9	Y	N
10	Y	
13	Y	

$$\begin{aligned} \text{info}_{\text{credit}}(D) &= \frac{8}{14} I(6,2) + \frac{6}{14} I(3,3) \\ &= \frac{8}{14} \left(-\frac{6}{8} \log_2 \left(\frac{6}{8}\right) - \frac{2}{8} \log_2 \left(\frac{2}{8}\right)\right) + \frac{6}{14} \left(-\frac{3}{6} \log_2 \left(\frac{3}{6}\right) - \frac{3}{6} \log_2 \left(\frac{3}{6}\right)\right) \\ &= 0.464 + 0.429 \\ &= 0.893 \end{aligned}$$

$$\begin{aligned} \text{Gain}_{\text{credit}} &= 0.940 - 0.893 \\ &= 0.047 \end{aligned}$$

สรุปได้ว่า Gain age แบ่งแยกข้อมูลได้ดีกว่า เมื่อเทียบกับค่ามากที่เลือก = 0.246